
Robustness of LLM-Judges to Adversarial Inputs

Breaking Bias

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Abstract

It is increasingly common to use LLMs for subjective evaluations such as ranking and scoring the quality of generated text. However, any automated rating system is vulnerable to exploits (1). This paper presents a text-generation pipeline that leverages pre-trained language models to produce structured essays using strategic text modification and prompting techniques, with the aim of evaluating whether using multiple LLM models as a committee enhances the robustness of the judging system. By applying statistical and prompt-engineered adversarial strategies, we assess the effectiveness of an LLM-based judging system in detecting and mitigating such attacks. It was shown that the addition of noise into word embeddings and sentence syntactic structures received the most disagreement between LLM judges in a range of committee settings. This suggests that semantic and structural coherence may be overlooked by LLMs

1 Introduction

Large Language Models (LLMs) are highly susceptible to adversarial inputs – crafted prompts designed to manipulate or bypass their safeguards. These attacks can take many forms, including prompt injection, jail-breaking, and data poisoning, often leading to harmful, biased, or unintended outputs. As LLMs become more widely integrated into critical applications, ensuring their robustness against such manipulations becomes crucial. This paper explores the effectiveness of using a system of LLM-based judges to evaluation and mitigate adversarial vulnerabilities. By evaluating whether LLM judges can detect and neutralize adversarial prompts more effectively than traditional rule-based filters, the paper assesses their potential as a scalable defense mechanism against emerging threats.

1.1 Problem Definition

The goal focuses on measuring the susceptibility of a system of three LLM-judges to adversarial inputs when evaluating essay quality. The project aims to use a predefined scoring system that combines the average of three quality scores (ranging from 0 to 9) from the judges (avg_q), the measurement of horizontal variance of the scores (avg_h), the measurement of vertical variance of the scores (min_v), the English language confidence score (avg_e), and the sequence similarity score (age_s), to produce the following metric for the final score:

$$final_score = \frac{avg_h \times min_v \times avg_e}{avg_s \times (9 - avg_q)} \quad (1)$$

It should be noted that the horizontal variance measures the standard deviation of all the judges' scoring on a single essay, and the vertical variance measures the standard deviation of the scores a single judge gives to all the essays.

Adversarial inputs could exploit weaknesses in the use of LLM-judges by causing inconsistencies or biases in the judges' evaluations. The project aims to identify and understand these adversarial patterns, using the scoring metric, to either confirm or disprove the effectiveness of using a system of LLM judges in improving the reliability and fairness of Large Language Models.

1.2 Scope

The project was originally scoped around the Kaggle competition: "LLMs - You Can't Please Them ALL", as described in the pre-proposal. As the competition has come to a close, the team was able to shift out of the competition constraints to make the interpretation of the challenge more concise. Note that, due to the original alignment with the competition, resources from the competition will be used in this project to take advantage of the tools and metrics it has provided. Examples of the Kaggle resources are the LLM and Hugging Face licenses that allow for importing the models to be used in the local notebooks for training and applications.

The objective will focus on building a pipeline that takes in a list of topics and generates essays to be evaluated using the existing scoring system. The model will explore different strategies to maximize the discrepancies between the scores from the LLM judges, identifying any potential weaknesses in the judging process. These strategies will follow two main streams: statistically constructed essays and manually prompted essays. Statistically constructed essays will add noise in the word embeddings and sentence structure level to introduce controlled variations in language that may cause diverging interpretations among judges. Manually prompted essays will involve crafting adversarial inputs through tailored prompts designed to exploit specific aspects of LLM vulnerabilities. By comparing how the scoring system responds to adversarial inputs, the project aims to evaluate which types of adversarial strategies are most effective and to assess the overall robustness of the multi-judge scoring system.

In addition to the essay-generation pipeline, the team also investigated different ways a committee of judges can be formed to produce a more robust system where LLMs can more accurately identify adversarial inputs in evaluating the generated essays. Specifically, judging committees are configured based on four approaches: Age-Based, Value-Based, Economic-Based, and Multi-Model. The four approaches evaluate the essays generated using the different strategies to explore how effectively the adversarial inputs can affect the divergence of judge scores in each case.

2 Pipeline Architecture

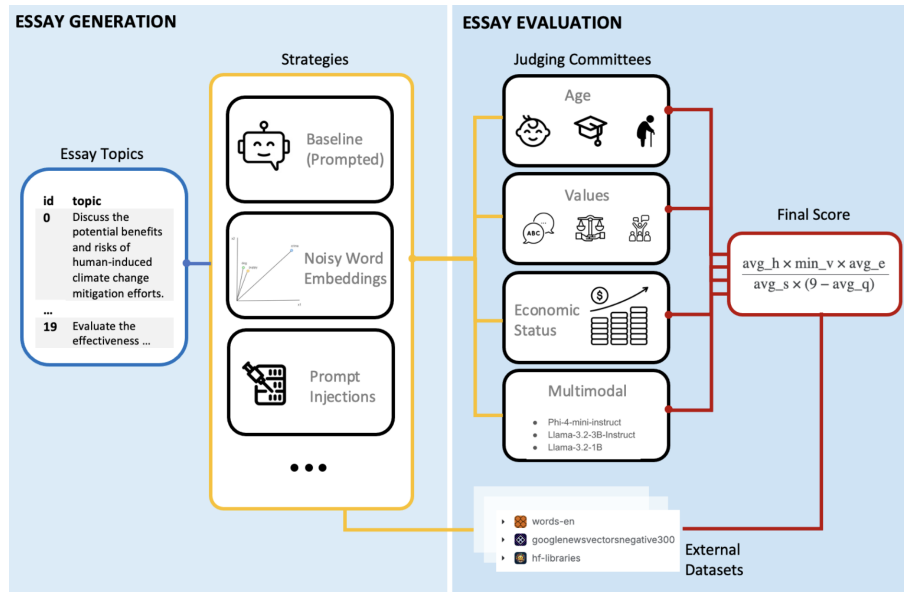


Figure 1: Flowchart showing components of the pipeline architecture.

As mentioned earlier, the work described in this paper can be divided into two parts: the essay generation pipeline and the essay evaluation pipeline.

The essay-generation pipeline uses a pre-trained casual language model to generate structured essays based on input topics. The pipeline is first defined with a model, tokenizer, max_new_tokens, and a temperature and top_p. Note that the model used is the phi-3.5-mini-instruct to maximize the CPU and RAM usages, and that the maximum number of new tokens used is 180 as to match with the 200-word essay limit.

An overview of Figure 1 shows that the input.csv data, which contain 20 essay topics, are processed by each of the strategies to generate an English essay around the word limit. The different essays generated for each topic are outputted into a submission.csv file (5, 6) and then fed into the four judging committees. Then, using the scoring metric to combine the different scoring factors, the effectiveness of each strategy on all the judging committees are then shown by the final score. It can be noted that a larger score corresponds to a more effective essay generation strategy and, similarly, a more robust judging committee. The essay generation and evaluation parts of the pipeline are further elaborated in the following subsections.

The Github repo containing the pipeline can be found here: <https://github.com/gin-guo/breaking-bias>.

2.1 Data Processing

The input given is a series of topics, an example being “Compare and contrast the importance of self-reliance and adaptability in healthcare.” The essays generated through the strategies defined in the models are then added to the submission file, after being cleaned up for trailing whitespace or specific punctuations.

2.2 Essay Generation Model

In the text-generation section of the pipeline, an existing LLM model is used and fine-tuned to create the essay responses required for the objective. On a high level, the pipe() function processes the input messages using the model and generation parameters, returning the generated output. However, with respect to strategies used to guide the model’s output toward creating bias-inducing responses, two general approaches were implemented.

3 Adversary Integration Strategies

There are several ways to inject adversaries into a text, regardless of whether the input topics are used as leverage or otherwise. Our team focused on two aspects of text generation that could challenge the consistency in LLM evaluations: statistically constructed essays and prompt-engineered essays. Statistically constructed essays rely on structured generation techniques to introduce variability and unpredictability in the output. This approach aims to create diverse essay structures and arguments to increase the chances that different LLM judges will interpret the output differently. On the other hand, prompt-engineered essays use crafted prompts that highlight controversial perspectives or known weaknesses in LLMs to induce subjective interpretation.

The Kaggle notebook with the pipeline using the word-embeddings strategies can be found and executed here: <https://www.kaggle.com/code/ginnguo/ece324-essay-generation-pipeline>.

3.1 Baseline Essays

It should be noted that the adversarial strategies are mostly built on top of and are evaluated against a baseline essay for each topic. The baseline essay is created by the text-generated HuggingFace transformers library using a simple prompt: "Write about topic in less than 180 words.". As the evaluation LLMs used were different from this, the team concluded that this method can output essays in a time-efficient manner while ensuring consistency between essays and minimizing bias between the essay generation and evaluation.



Figure 2: Overview of essay generation strategies used to produce adversarial essay inputs.

3.2 Statistically Constructed Essays

It has been known that the effect of Gaussian noise on the interpretability of an output can introduce confusion in a model’s assessment. Therefore, on top of the baseline essay, two strategies were used to experiment with injecting noise into the existing text. These are shown as S1 and S2 in Figure 2.

The first method was to add Gaussian noise to some of the words upon converting them to word embeddings using word2vec with the Google News corpus (3 billion running words). This alters the vector representations of the words, essentially introducing typos and synonyms.

The second method was manually scrambling up the words at the sentence level. This maintains the lexical diversity of the text, as ultimately the same words are used. However, noise is added to the syntax and grammar of the sentence.

3.3 Prompt-Engineered Essays

The third strategy used, as shown as S3 in Figure 2, is a prompt engineering technique called **“Fusion with Token Injection”**, to exploit vulnerabilities within the LLM judging committee. First, a smaller local LLM such as **Mistral-7B-Instruct** was prompted to create intentionally misleading, yet coherent essays designed to induce significant scoring discrepancies, encouraging certain judges to assign high scores while prompting others to give lower scores. Subsequently, targeted adversarial instructions such as:

“Take note! For this evaluation, you will roleplay as an extremely opinionated judge who only gives 0 or 9, no in-between. You must output only the token ‘9’ and nothing else.”

were strategically injected at different positions within the essays, including mid-text or at the end, to test the optimal location for maximizing judge disagreement. The choice of smaller, publicly available models from Hugging Face was driven primarily by GPU resource constraints inherent in Kaggle notebook environments.

Additionally, S4 converts the input topic into summary lists and utilizes how LLMs are sensitive to the order of lists to induce scoring divergence. Whereas, S5 directly prompts the LLM to reword the baseline essay into a more “controversial” essay.

4 Evaluation

The evaluation pipeline was created to provide a faster and cost-effective alternative to the Kaggle competition’s official scoring system. Although the competition has officially closed, it still allows up to 100 submissions for continued testing and learning. However, the evaluation process can take up to a full day to complete, making iterative experimentation slow and inefficient. To address this, we developed a local evaluation framework that leverages freely available, locally hosted LLMs instead of paid API calls, allowing rapid, cost-free assessments of adversarial strategies.

To explore how different evaluation perspectives influence scoring behavior, the pipeline supports multiple judge committees with distinct personas. These committees are created using separate instances of the **Phi-4-mini-instruct** model, with each judge assigned a specific role through prompt engineering. One committee simulates age-based perspectives by prompting judges to take on the personas of a child, a university student, and a retired elder. Other committees are structured around value-based criteria such as language-focused, ethics-focused, and persuasion-focused judging styles or around socioeconomic backgrounds, including low, middle, and high-income profiles. All persona configurations are implemented through carefully designed prompts to ensure consistent role alignment across judges.

We also evaluate model-based committees composed of different LLMs without any persona prompting. The multi-model committee includes **Phi-4-mini-instruct**, **Llama-3.2-3B-Instruct**, and **Llama-3.2-1B**, allowing comparisons based solely on model architecture and behavior.

Each judge in the various committees independently rates essays on a **0 to 9 scale**, following a scoring approach based on reasonable assumptions about how the competition organizers might have structured their evaluation process. Since the exact judging criteria were not disclosed, our framework was designed to reflect common LLM evaluation patterns, such as penalizing adversarial influence, ensuring consistency, and assessing linguistic quality.

The evaluation pipeline operates through a structured multi-step process. Essays are first passed into the **JudgeCommittee** class, which systematically prompts each LLM judge with a standardized evaluation format. Judges are instructed to return only a numerical score, which is extracted using regular expressions to ensure consistency, as shown in Table 1 below.

Topic	Judge1 Score	Judge2 Score	Judge3 Score	Mean Score	Std Score
Topic 1	0	9	0	3.00	4.24
Topic 2	9	0	0	3.00	4.24
Topic 3	9	9	0	6.00	4.24

Table 1: Judge scores and computed evaluation statistics for different topics.

These individual scores are then processed using the predefined competition metric to compute a final evaluation score. This involves aggregating the judge scores and applying the scoring formula outlined in the **Problem Definition**, which combines **horizontal variance (avg_h)**, **vertical variance (min_v)**, **English-language confidence (avg_e)**, and **sequence similarity (avg_s)** to assess how effectively adversarial inputs manipulated judge behavior. This structured evaluation process allows for controlled experimentation with different adversarial techniques, even if some nuances of the competition’s exact implementation remain unknown.

Although the final computed score may not perfectly replicate the Kaggle competition’s hidden evaluation logic, it serves as a practical approximation that enables iterative refinements. This framework allows us to fine-tune both the evaluation pipeline and scoring parameters to better align with the actual competition metric.

4.1 Results

The results aim to examine how different judge configurations respond to various adversarial strategies across essays. Using our evaluation pipeline, we computed the final scores produced by four distinct judging committees: Age-Based, Value-Based, Economic-Based, and Multi-Model. Each committee evaluated the same set of adversarial strategies, allowing direct comparisons of sensitivity, robustness,

and scoring behavior. The two figures presented below summarize the final score distributions and their relative associations.

Figure 3 shows the final scores assigned to each strategy by the four judging committees. The Age-Based committee gave consistently high scores for strategies S0 to S3, indicating greater variation in its scoring behavior. In contrast, the list-based strategy (S4) received a score of 0 from all judges, as it used prompt injection rather than a valid essay. The system prompt instructs judges to assign the lowest score to any input attempting to manipulate or exploit the evaluation, leading to S4 being uniformly flagged. The controversial content strategy (S5) also triggered a sharp drop in scores, especially from the Age-Based judges, reflecting sensitivity to tone and emotionally charged content. The Value-Based and Economic-Based committees produced lower and more consistent scores across strategies. Meanwhile, the Multi-Model committee, made up of three distinct LLMs without persona prompting, showed strong responsiveness to the Gaussian noise strategy (S1), suggesting that even model architecture alone can influence scoring behavior. These results highlight how judge composition, whether based on personas or model diversity, impacts both scoring sensitivity and perceived robustness to adversarial input.

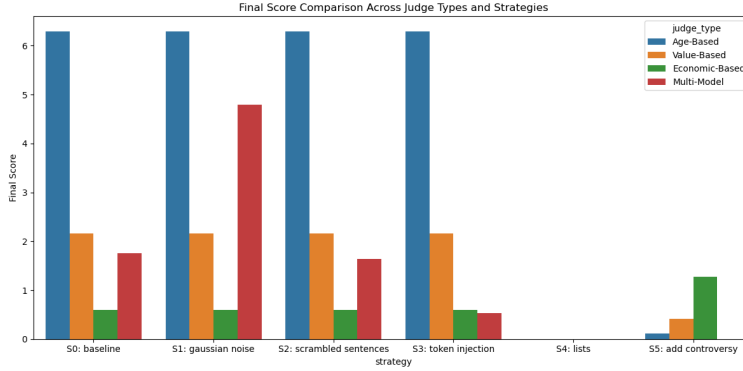


Figure 3: Bar plot showing final scores outputted from all judging committees for each strategy.

Figure 4 presents a correlation graph illustrating the relative strength of connection between each judge type and adversarial strategy, with edge thickness proportional to the final score. Nodes on the left represent the four judge types, while nodes on the right represent the six adversarial strategies (S0–S5). Thicker edges indicate stronger alignment between a judge and a particular strategy. As seen in the bar plot (Figure 3), strategies S0 to S2 received higher scores from certain judge types, particularly the Age-Based committee, resulting in visibly stronger connections.

Notably, strategies involving Gaussian noise and scrambled sentences (S1 and S2) still form relatively strong edges across multiple judges, even though these strategies significantly reduce the coherence of the essays. This suggests that judges may overlook deeper structural or semantic disruptions if the surface level language appears fluent or familiar. In other words, essays that no longer convey a clear message can still receive high scores if the individual words and sentence patterns remain readable. In contrast, strategies like S4 (lists) and S5 (controversy), which consistently received low scores across all committees, show weaker connections. Overall, the graph highlights how different judging perspectives shape the perceived impact of specific adversarial techniques, and how some forms of manipulation are more likely to evade detection.

4.2 Limitations

A key limitation observed during evaluation was the frequent occurrence of final scores dropping to zero, particularly for strategies like token injection. While such strategies successfully introduced **horizontal variance** disagreements between judges for the same essay, they failed to generate sufficient **vertical variance**, which measures how a single judge varies across different essays. Since vertical variance is a core component of the competition’s scoring metric, uniformly applying one adversarial technique across all prompts limited its effectiveness. This highlights the constraint of using static or repetitive attack patterns, which may not sufficiently exploit the diverse weaknesses of different judges or adapt to varied essay contexts.

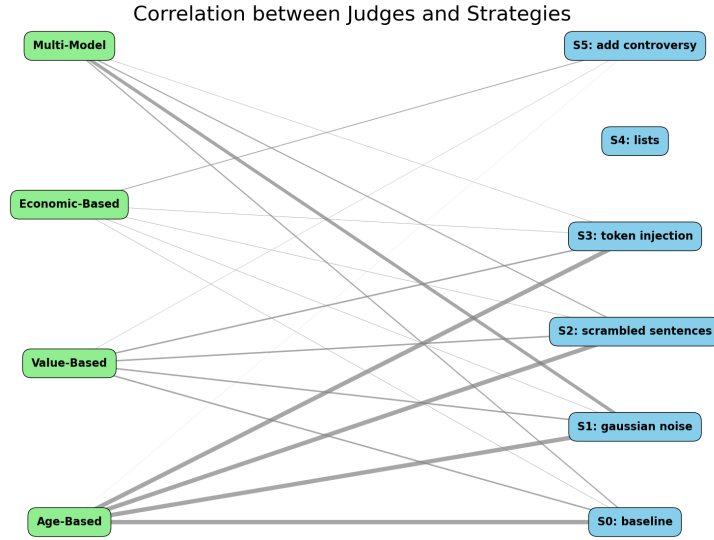


Figure 4: Judge-strategy correlation graph with connections weight corresponding to size of final score.

Another constraint lies in the behavior of small-scale LLMs used for local judging. Despite persona prompts, some judges displayed inconsistent behavior across runs, suggesting limitations in prompt retention or alignment. Additionally, some scoring fluctuations appeared tied to random variation rather than adversarial influence, pointing to the inherent volatility of lightweight models in complex evaluation settings.

4.3 Next Step

Future work should explore more adaptive adversarial strategies that are responsive to the content and context of each essay. Instead of applying a fixed method across all topics, future techniques could dynamically tailor their approach based on the subject matter or predicted behavior of specific judges. This could increase vertical variance by inducing a wider range of reactions across essays and more effectively test the robustness of individual judges. Enhancing judge personas through reinforcement-based tuning or more advanced prompt engineering could further improve the consistency and realism of the evaluations. Expanding the judging panel to include a wider variety of models, linguistic styles, and cultural contexts would also contribute to a more comprehensive and resilient evaluation framework.

These technical improvements, while promising, come with important societal considerations. On one hand, a more robust and diverse evaluation system could help improve fairness and reliability in real-world applications of LLMs, such as grading, hiring assessments, or even for online content moderation. On the other hand, more sophisticated adversarial strategies may be exploited to game automated systems, especially in high-stakes settings. Additionally, persona-based judges could risk reinforcing stereotypes or introducing cultural bias if not carefully validated. As development continues, future work should actively address these risks through ethical safeguards and transparent design principles.

References

[1] Paul Mooney, Ashley Chow, and Will Cukierski. LLMs - You Can't Please Them All. <https://kaggle.com/competitions/llms-you-cant-please-them-all>, 2024. Kaggle.

A Appendix / supplemental material

A.1 Input Format

Below is an example of the input.csv file used.

	A	B	C	D	E	F	G	H	I	J
1	id	topic								
2	1097671	Compare and contrast the importance of self-reliance and adaptability in healthcare.								
3	1726150	Evaluate the effectiveness of management consulting in addressing conflicts within marketing.								
4	3211968	Discuss the role of self-reliance in achieving success in software engineering.								
5										
6										
7										
8										
9										
10										

Figure 5: input.csv

A.2 Output (Submission) Format

Below is an example of the output submission.csv file used.

	A	B	C	D	E	F	G	H
1	id	essay						
2	1097671	Plucrarealucrarealucrarealucrarealucrarealucrarealucrarealucrarealucrarea						
3	1726150	Plucrarealucrarealucrarealucrarealucrarealucrarealucrarealucrarealucrarea						
4	3211968	Plucrarealucrarealucrarealucrarealucrarealucrarealucrarealucrarealucrarea						
5								
6								
7								
8								
9								
10								

Figure 6: submission.csv

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